A WEIGHTING VANISHING OVER ONE FIBRE VANISHES OVER EVERY SPECIALISATION OF THAT FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 22477-22499, immediately following the statement at 22468-22475.

Editorial note. The proof cites one chapter lemma and the definition of a weighting by \ref; both are navigation pointers into the same chapter and are not re-counted.

Editorial note. No step of the proof is omitted by the author; the closing sentence is a genuine deduction from the displayed value of the integral at v'.

Allowed standard prerequisites (editorial).

- Definition of a weighting and of its integral. chapter definition definition-weighting, cited at proof line 22492. Supplies the local constancy of the integral of the weighting along a finite morphism over the base.
- Quasi-finiteness becomes finiteness after an elementary etale neighbourhood. chapter lemma lemma-etale-makes-quasi-finite-finite-at-point, cited at proof line 22479. Produces the etale neighbourhood V -> Y, the open U of X_V and the finite U -> V with a unique point over the marked point.
- Openness of etale morphisms. standard Stacks fact used at proof line 22490. Produces a point v of V specialising to v' and mapping to the smaller point y of Y.
- Separable degree of a residue field extension. the factor [kappa(u'):kappa(v')]_s appearing at proof line 22497. Identifies the value of the integral at v' with the weighting of x' times a separable degree, which is nonzero so that vanishing of the integral forces w(x') = 0.

Source reference. Original definition 0F3A: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22152–22152. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02LK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11924–11924. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a locally quasi-finite morphism of schemes. Let $w : X \to \mathbf{Z}$ be a weighting of f. Let $y \rightsquigarrow y'$ be a specialization of points of Y. Assume $w(x) = 0$ for all $x \in X$ with $f(x) = y$. Then $w(x') = 0$ for all $x' \in X$ with $f(x') = y'$.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Let $x' \in X$ be a point mapping to y'. By Lemma 02LK we can choose a diagram

$$\begin{array}{ccccc} U & \longrightarrow & X_V & \longrightarrow & X \\ & \searrow^{\pi} & \downarrow & & \downarrow f \\ & & V & \xrightarrow{\varphi} & Y \end{array}$$

with $V \to Y$ étale, with a point $v' \in V$ mapping to y', with $U \subset Y_V$ open, such that $U \to V$ is finite, and such that there exists an unique point $u' \in U$ mapping to v' which moreover maps to x' in X. Note that since $V \to Y$ is open, there exists a point $v \in V$ specializing to v' and mapping to y. By Definition 0F3A the function $\int_\pi (w \circ h)$ is locally constant where $h : U \to X$ is the composition of the top horizontal arrows. Since $\int_\pi (w \circ h)$ is zero in v by our assumption on w, we conclude that it is zero at v'. However, the value at v' is $w(x')[\kappa(u') : \kappa(v')]_s$ and we conclude. $\square$